PRITOM MAJUMDER

Email | LinkedIn | Research Portfolio | ResearchGate | +880 1859 472796 | Dhaka, Bangladesh

RESEARCH INTEREST

Functional, Piezoelectric Ceramics & Nanomaterials; Oxide Semiconductor Thin Films; Photonic & Optoelectronic Materials; Polymer Composites & Nanocomposites; Metallurgy; Computational Material Science (DFT).

EDUCATION

B.Sc. in Materials and Metallurgical Engineering **2021 to June 2026**

Bangladesh University of Engineering and Technology (BUET), Dhaka

CGPA: 3.64 (15^{th} of 59) / 4.00

Relevant coursework: Advanced Ceramics (A+); Nanostructured Materials & Thin Films (A+); Electrical, Optical and Magnetic Properties of Materials (A+); Characterisation of Materials Laboratory (A+); Corrosion and Degradation of Metals and Alloys (A+); Surface Engineering and Tribology (A+); Photonic and Magnetic Materials (A).

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher · Ahmed Sharif's Lab · Department of MME, BUET **2025 to 2026**

"Investigation of Dielectric and Piezoelectric Properties of KNN with BCHfT Composite Ceramics" ; Supervisor: Prof. Ahmed Sharif ; THESIS BOOK & THESIS PPTX

- Synthesized lead-free $K_{0.5}Na_{0.5}NbO_3$ (KNN) ceramics modified with $Ba_{0.85}Ca_{0.15}Hf_{0.1}Ti_{0.9}O_3$ (BCHfT) by the conventional solid-state route.
- Characterized phase evolution and microstructure by XRD and FE-SEM; measured frequency- and temperature-dependent dielectric permittivity and loss, impedance spectroscopy, and the piezoelectric charge coefficient d_{33}.
- Identified KNN-1wt% BCHfT as the optimum composition, giving the best balance of dielectric response.

Undergraduate Researcher · Solid State Physics Laboratory · Department of Physics, BUET **2024 to 2026**

- Synthesis of SnO2 and Cu:SnO2 research by hydrothermal process.

Undergraduate Thesis Researcher · Pilot plant and Process Development Center · BCSIR **2025 to 2026**

- Carried out the complete characterization of the KNN-BCHfT thesis series on BCSIR instruments: XRD, SEM, frequency- and temperature-dependent dielectric test.

SELECTED PROJECTS

Cu-Doped SnO_2 Thin Films · Solid State Physics Laboratory, BUET **2024 to 2026**

Hydrothermal growth and defect engineering ; Supervisor: Dr. Md. Azizar Rahman ; Manuscript in preparation (first author)

- Targeted p-type Cu:SnO_2 for an all-SnO_2 p-n homojunction diode, a route that collaborator-run DFT+U calculations indicated would require the oxygen vacancies to be suppressed.
- Grew undoped and Cu-doped (5, 10 and 15 at%) SnO_2 thin films directly on soda-lime glass by a single-step hydrothermal route at 180 °C, using NaOH and NaCl as growth mediators to tailor the defect chemistry during nucleation.
- Stabilized the metastable orthorhombic (α-PbO_2-type) polymorph at ambient pressure, confirmed by XRD, HRTEM, SAED and EDS. Tracked the doping-driven band gap by UV-Vis.
- Established from I-V and Mott-Schottky analysis of the Cu:SnO_2/ITO junction that Cu doping does not invert the conductivity type (remained n-type).

Millsorpt - Capstone Design Project · Department of MME, BUET **2025**

- Converted ferrous mill scale, a low-value steel-industry waste, into a chemically treated adsorbent (hematite rich iron-oxide) for heavy metal wastewater treatment
- Designed a low-cost bench scale filtration column achieving a high removal of Cu, Cr and Ni from mixed-metal solutions; iron-oxide phases confirmed by XRD, metal removal quantified by AAS and UV-Vis.
- Presented the work to specialists at Abul Khair Steel, BCSIR, the Bangladesh Atomic Energy Commission etc., receiving positive technical feedback.

Polymer-Matrix Composite for a Car Clutch Disc · Materials Testing Laboratory, BUET · Group **2024**

- Designed and fabricated a fiber-reinforced asbestos–epoxy composite replica of a car clutch disc by hand layup
- Tested the cured composite by tensile, Shore-D hardness, impact and three-point flexural testing, and identified the matrix polymer from its FTIR spectrum.
- Compared the product with commercial clutch facings.

3D PotterBot - CAD Modelling · SolidWorks **2023**

- Modelled a paste-extrusion ceramic 3D printer as a fully-mated 118 parts assembly; produced photorealistic renders, an exploded-view motion study, and a static Finite Element Analysis of the vertical support rod.

ON GOING WORK

1. **Pritom Majumder**, Anabil Dey, and Ahmed Sharif. "Advancements in Cold Sintering of Piezoceramics: Flux Mediation, Microstructural Evolution, and Enhanced Dielectric Properties - A Review." (under peer review) Journal: Emergent Materials.

INDUSTRIAL AND LABORATORY TRAINING

Bangladesh Council of Scientific and Industrial Research (BCSIR) · Dhaka **2026**

Abul Khair Steel Melting Limited (AKSML) · Chittagong **March 2025**

Bangladesh Industrial Technical Assistance Centre (BITAC) · Dhaka **March 2025**

TECHNICAL SKILLS

- **Synthesis, Processing and Data Analysis:** Solid-state ceramic synthesis; hydrothermal synthesis; sintering; uniaxial pressing; XRD; FE-SEM and SEM; TEM including HRTEM, SAED; Impedance spectroscopy; frequency- and temperature-dependent dielectric measurement; d_{33} meter; cyclic voltammetry; I-V and Mott-Schottky analysis; UV-Vis spectroscopy; photoluminescence (PL); FTIR
- **Mechanical & Chemical Testing:** Tensile, Charpy impact and three-point flexural testing; Vickers, Rockwell and Brinell hardness; optical emission spectroscopy (OES); gravimetric and wet chemical analysis
- **Software:** OriginPro; MATLAB; SolidWorks; ImageJ; X'Pert HighScore; C/C++; MS Office; Adobe Photoshop and Illustrator

AWARDS AND SCHOLARSHIPS

- **University Merit Scholarship**, Bangladesh University of Engineering and Technology **(July 2022)**
- **University Stipend Scholarship**, Bangladesh University of Engineering and Technology (July 2021 and January 2022)
- **Letter of Appreciation**, Abul Khair Steel Melting Limited - for professionalism and work ethic during industrial training **2025**
- **Certificate of Perfect Attendance**, Notre Dame College, Dhaka **(2019 to 2020)**

LANGUAGE PROFICIENCY

English: IELTS Academic - Overall Band 7.0 (Listening 7.5, Reading 7.5, Writing 6.0, Speaking 6.5) **2026**

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

- **Director of Administration**, Executive Committee, BUET Debating Club (BUETDC), 2025-26 :- previously Information Secretary. Organizing panel member for BUETDC Nationals 2024; finalist at the BUET Freshers' Debating Tournament; competed at the Inter-Faculty Tournament 2024 and the 6th GSCDC Nationals 2024.
- **Organizer**, Winter Fitness Competition 2025, BUET Gymnasium :- planned and ran the event in collaboration with the resident trainer; Organized Treasure-Hunt competition 2024, as an organizing panel member of "Reception'21"
- **Class Representative**, Department of MME, BUET, 2021-22 :- elected by classmates for both semesters of the first year
- **Active Member & Blood Donor**, Badhon - a non-profit voluntary blood donation organization, 2022-present